

One says moving.
One says
it never moved.

GPS said the car was
travelling. The
accelerometer had not
twitched in ten minutes.

Only one of them is right,
and it is not the one you
think.



// THE CAST

Cannot say where.
Knows whether.



it moved

THE SECOND WITNESS

exhibit 02

`Sensor.TYPE_ACCELEROMETER`

Useless on its own. Decisive standing next to a witness with a concussion. It fails differently, which is the entire reason it is worth having.



The Loopdown

@darkpandawarrior

02 / 05

swipe

// THE BETTER QUESTION

Rank per question, not overall

```
fun process(  
    fix: GpsFix,  
    motionStill: Boolean = false,    // IMU says physically still  
    harshAccel: Boolean = false,    // hard accel or braking this fix  
): ProcessResult?
```



Corroborate, do not tune

01 Find a signal that fails differently
a second thermometer is not a second opinion

02 Ask what each is authoritative for
IMU owns "moving". GPS owns "where".

03 Disagreement is information
log it, do not average it away

04 Stop tuning the lonely sensor
a filter on one input is a guess about itself



Correlation beats cleverness

Do not build a smarter threshold on a lonely sensor.

Find a second signal that fails differently, and give it authority over the one question it is actually good at.

